

Question				Marking details	Marks available														
					AO1	AO2	AO3	Total	Maths	Prac									
9/1	(a)			Change in {Genes/DNA/chromosomes/ genetic material/ base sequence/ genetic make up} Reject different DNA	1			1											
	(b)			(Ionising) radiation/ UV light/ x rays/ gamma rays/ carcinogens	1			1											
	(c)	(i)		Gametes correct (1) Offspring genotypes correct (1) <table border="1"><tr><td></td><td>N</td><td>n</td></tr><tr><td>N</td><td>NN</td><td>Nn</td></tr><tr><td>n</td><td>Nn</td><td>nn</td></tr></table> If gametes incorrect 0 marks. Allow 1 ECF mark for genotypes If incorrect letters used 0 marks for gametes 1 ECF mark for genotypes. If X, Y used 0 marks for Punnett square		N	n	N	NN	Nn	n	Nn	nn		1 1		2		
	N	n																	
N	NN	Nn																	
n	Nn	nn																	
		(ii)		25%/ ¼ / 0.25 Reject any ratios/ 25 unqualified/ 2/8 Ecf from (i) if relates to homozygous recessive genotype in Punnett square			1	1											
	(d)	(i)		Gene therapy/ somatic cell therapy Reject gene treatment	1			1											
		(ii)		Aerosol/inhaler Reject asthma inhaler/ inhaling	1			1											

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		(iii)		Any one (×1) from: Has to be repeated / is not permanent/ only temporary/ short term/ not a cure (1) Genes not passed on (1) Low numbers of cells take up gene (1) difficulty in targeting appropriate cells (1)	1			1		
				Question 9/1 total	5	2	1	8	0	0